

***Generation Interconnection
System Impact Study Report***

For

***PJM Generation Interconnection Request
Queue Position AD1-020***

Hunterstown - Lincoln 115 kV

(Revised)

December 2018

Preface

The intent of the System Impact Study is to determine a plan, with approximate cost and construction time estimates, to connect the subject generation interconnection project to the PJM network at a location specified by the Interconnection Customer. As a requirement for interconnection, the Interconnection Customer may be responsible for the cost of constructing: Network Upgrades, which are facility additions, or upgrades to existing facilities, that are needed to maintain the reliability of the PJM system. All facilities required for interconnection of a generation interconnection project must be designed to meet the technical specifications (on PJM web site) for the appropriate transmission owner.

In some instances an Interconnection Customer may not be responsible for 100% of the identified network upgrade cost because other transmission network uses, e.g. another generation interconnection or merchant transmission upgrade, may also contribute to the need for the same network reinforcement. The possibility of sharing the reinforcement costs with other projects may be identified in the Feasibility Study, but the actual allocation will be deferred until the System Impact Study is performed.

The System Impact Study estimates do not include the feasibility, cost, or time required to obtain property rights and permits for construction of the required facilities. The project developer is responsible for the right of way, real estate, and construction permit issues. For properties currently owned by Transmission Owners, the costs may be included in the study.

General

Adams Solar (“Interconnection Customer”) has proposed a new solar and battery storage generation facility located approximately one mile from Hunterstown substation in Straban Township, Adams County, Pennsylvania. The installed facilities will have capability of 100 MW with 53.6 MW of this output being recognized by PJM as capacity.

The following table shows the MW systems contributions of each fuel type as related to the Maximum Facility Output and Capacity Interconnection Rights (CIR):

Solar	Battery	Total MW
80	20	100 (MFO)
53.6	0.0	54.6 (CIR)

The proposed in-service date is September 30, 2020. This study does not imply a **Mid-Atlantic Interstate Transmission (“MAIT” or “Transmission Owner”)** commitment to this in-service date.

Point of Interconnection (POI)

This project is tapping the Hunterstown – Lincoln 115 kV transmission line. The Point of Interconnection for this project will be accomplished by constructing a new 115 kV three (3) breaker ring bus station and looping the Hunterstown – Lincoln 115 kV line into the new station. The substation will be located approximately 1.0 miles from Hunterstown substation. Please see Appendix 2 for a one-line diagram of this interconnection.

Summary of Estimated Costs and Transmission Owner Scope of Work

The following upgrades are required to support AD1-020 Primary Point of Interconnection; all associated PJM network upgrade numbers will be determined at the facilities study stage. Below shown costs do not include Contribution in Aid of Construction (CIAC) Federal Income Tax Gross Up charge. The total tax dollars amount (\$ 1,257,100) may or may not be charged to this project depending upon whether this project meets the eligibility requirements of the latest IRS Safe Harbor provisions for non-taxable status.

(a) Attachment Facilities: None.

(b) Direct Connection Network Upgrades:

- (b1) Install new 115kV three-breaker ring bus substation\$ 3,876,400
- (b2) Loop the 962 (Hunterstown-Lincoln) 115kV circuit into substation.....\$ 466,700
- (b3) Project management, commissioning, environmental, real estate, and SCADA.\$ 339,900
- (b4) Revenue metering - engineering oversight of specification and design of new revenue metering that will be installed by power producer (interconnection customer) at their location (AD1-020) and connected to the new ring bus station on the Hunterstown - Lincoln line. Coordinate FE MV90 access to the new meter.....\$ 2000

(c) Non-Direct Network Upgrades:

- (c1) Replace one (1) existing shield wire with optical ground wire (OPGW) on the Hunterstown-Lincoln 115kV circuit between the proposed AD1-020 ring bus and Lincoln substation, approximately 1.6 miles.\$ 496,600
- (c2) Replace one (1) existing shield wire with OPGW on the Hunterstown-Lincoln 115kV circuit between the proposed AD1-020 ring bus and Hunterstown substation, approximately 1.0 miles.\$ 321,300
- (c3) Install new line relaying and current-voltage transformers (CVT) for the AD1-020 Interconnection at Hunterstown substation.\$ 259,200
- (c4) Install new line relaying and current- voltage transformers (CVT) for the AD1-020 Interconnection at Lincoln substation.....\$ 259,200
- (c5) Estimated MPLS router at new AD1-020 Interconnection Substation to support new RTU.....\$ 150,000

(d) Direct Local Network Upgrades: None.

(e) Non-Direct Local Network Upgrades: None.

(f) Option to Build Upgrades: None.

Estimated Total Costs (a) to (f):\$ 6,171,500

Interconnection Customer Requirements and Responsibilities and Compliance Related Matters

1. The proposed Customer Facilities must be designed in accordance with FE's "Requirements for Transmission Connected Facilities" document located at:
<http://www.pjm.com/planning/design-engineering/to-tech-standards/private-firstenergy.aspx>. In particular, Interconnection Customer is responsible for the following:
2. The purchase and installation of fully rated 115 kV circuit breaker to protect the AD1-020 generator lead line. A single circuit breaker must be used to protect this line; if the project has several GSU transformers, the use of the individual GSU transformer breakers to protect the line is not acceptable by FirstEnergy.
3. The purchase and installation of the minimum required FirstEnergy generation interconnection relaying and control facilities. This includes over/under voltage protection, over/under frequency protection, and zero sequence voltage protection relays.
4. The purchase and installation of supervisory control and data acquisition (SCADA) equipment to provide information in a compatible format to the FirstEnergy Transmission System Control Center.
5. The compliance with the FirstEnergy and PJM generator power factor and voltage control requirements.
6. Power Factor Requirements: Interconnection Customer shall design its solar/battery powered non-synchronous Customer Facility with the ability to maintain a power factor of at least 0.95 leading (absorbing VARs) to 0.95 lagging (supplying VARs) measured at the high-side of the facility substation transformer(s) connect to the FirstEnergy transmission system.
7. The execution of a back-up service agreement to serve the customer load supplied from the AD1-020 generation project metering point when the units are out-of-service. This assumes the intent of Interconnection Customer is to net the generation with the load.
8. Interconnection Customer must meet all PJM, ReliabilityFirst and NERC reliability criteria and operating procedures required for standards compliance. For example, Interconnection Customer will need to properly locate and report the over and under-voltage and over and under-frequency system protection elements for its units as well as the submission of the generator model and protection data required to satisfy the PJM and ReliabilityFirst audits. Failure to comply with these requirements may result in a disconnection of service if the violation is found to compromise the reliability of the FirstEnergy system.
9. Interconnection Customer will be responsible for acquiring all easements, properties, and permits that may be required to construct both the new interconnection switching station and the associated attachment facilities. In addition, Interconnection Customer will also be responsible for the rough grade of the property and an access road to the proposed three breaker ring bus site. The project will also require non-direct connection upgrades at Hunterstown and Lincoln substations.
10. System Protection: The IC must design its Customer Facilities in accordance with all applicable standards, including the standards in FE's "Requirements for Transmission Connected Facilities" document located at above shown link. Preliminary Protection requirements will be provided as part of the facilities study. Detailed protection requirements will be provided once the project enters the construction phase.

The above requirements are in addition to any metering or other requirements of PJM.

Revenue Metering and SCADA Requirements

PJM Requirements

The Interconnection Customer will be required to install equipment necessary to provide Revenue Metering (KWH, KVARH) and real time data (KW, KVAR) for Interconnection Customer's generating Resource. See PJM Manuals M-01 and M-14D, and PJM Tariff Sections 24.1 and 24.2.

Interconnected Transmission Owner Requirements

The Interconnection Customer will be required to comply with all FirstEnergy Revenue Metering Requirements for generation interconnection customers which can be found in FirstEnergy's "Requirements for Transmission Connected Facilities" document located at:

<http://www.pjm.com/planning/design-engineering/to-tech-standards.aspx>

Schedule

Based on the scope of the direct connection for the Primary POI, it is expected to take a minimum of 23 months from the signing of a Connection Service Agreement to complete the installation required for the Project. This includes a preliminary payment that compensates FirstEnergy for the first three months of the engineering design work that is related to the construction of the AD1-020 115 kV interconnection substation. It also assumes that the Interconnection Customer will provide the property for the AD1-020 115 kV interconnection substation and all right-of-way, permits, easements, etc. that will be needed. A further assumption is that there will be no environmental issues with any of the new properties associated with this project, that there will be no delays in acquiring the necessary permits for implementing the defined direct connection and network upgrades, and that PJM will allow all transmission system outages when requested.

Network Impacts

The Queue Project AD1-020 was evaluated as a 101.0 MW (Capacity 53.6 MW) injection as a tapped connection into Hunterstown-Lincoln 115kV substation in the METED area. Project AD1-020 was evaluated for compliance with applicable reliability planning criteria (PJM, NERC, NERC Regional Reliability Councils, and Transmission Owners). Project AD1-020 was studied with a commercial probability of 100%. Potential network impacts were as follows:

Summer Peak Analysis - 2021

Generator Deliverability

(Single or N-1 contingencies for the Capacity portion only of the interconnection)

None

Multiple Facility Contingency

(Double Circuit Tower Line, Fault with a Stuck Breaker, and Bus Fault contingencies for the full energy output)

None

Contribution to Previously Identified Overloads

(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)

None

Steady-State Voltage Requirements

None

Short Circuit

None

Affected System Analysis & Mitigation

Delivery of Energy Portion of Interconnection Request

PJM also studied the delivery of the energy portion of this interconnection request. Any problems identified below are likely to result in operational restrictions to the project under study. The developer can proceed with network upgrades to eliminate the operational restriction at their discretion by submitting a Merchant Transmission Interconnection request.

Only the most severely overloaded conditions are listed. There is no guarantee of full delivery of energy for this project by fixing only the conditions listed in this section. With a Transmission Interconnection Request, a subsequent analysis will be performed, which will study all overload conditions associated with the overloaded element(s) identified.

None

Light Load Analysis - 2021

Not Required

System Reinforcements

Short Circuit

None

Stability and Reactive Power Requirement

Not Required

Summer Peak Load Flow Analysis Reinforcements

New System Reinforcements

(Upgrades required to mitigate reliability criteria violations, i.e. Network Impacts, initially caused by the addition of this project generation)

None

Contribution to Previously Identified System Reinforcements

(Overloads initially caused by prior Queue positions with additional contribution to overloading by this project. This project may have a % allocation cost responsibility which will be calculated and reported for the Impact Study)

None

Light Load Load Flow Analysis Reinforcements

New System Reinforcements

(Upgrades required to mitigate reliability criteria violations, i.e. Network Impacts, initially caused by the addition of this project generation)

None

Contribution to Previously Identified System Reinforcements

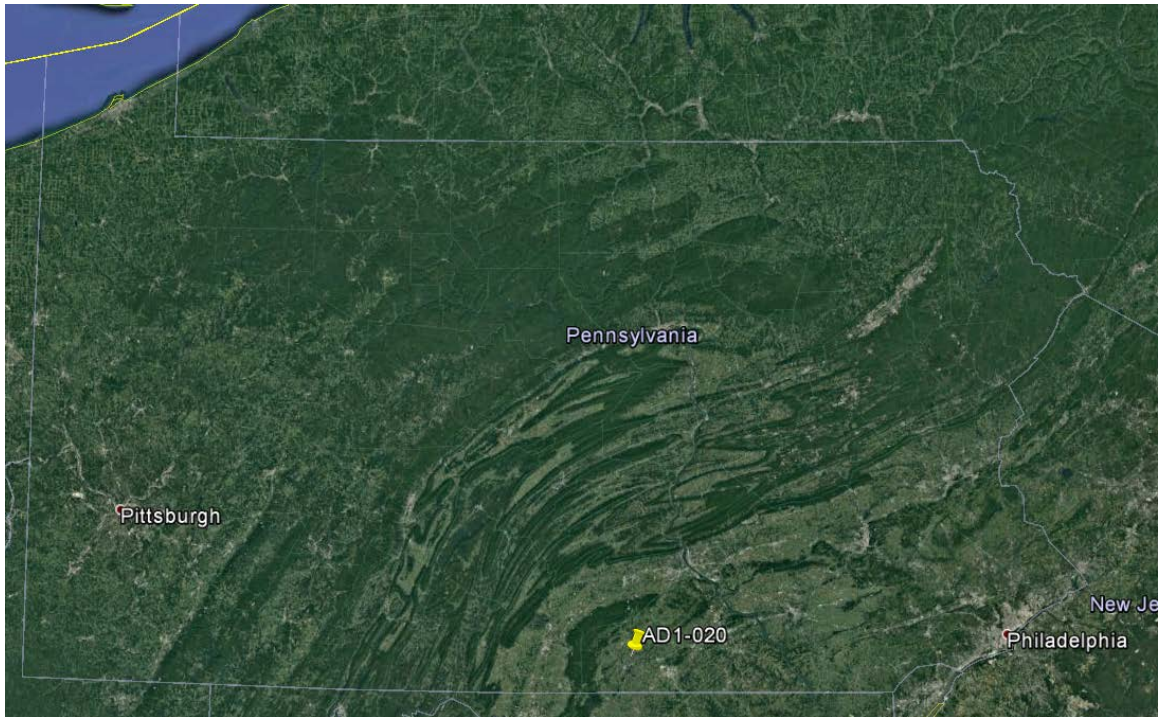
(Overloads initially caused by prior Queue positions with additional contribution to overloading by this project. This project may have a % allocation cost responsibility which will be calculated and reported for the Impact Study)

None

Appendix 1

Facility Location

PJM Queue Position: AD1-020



Interconnection One-Line Diagram
PJM Queue Position: AD1-020

